Interactions Between Pruning, Quantization, and Batch Size in MLPs

Hema
IIT Madras Zanzibar
zda24b025@iitmz.ac.in

Abstract

Neural networks are usually larger than they need to be, which makes them slow to run and heavy to store. Compression methods such as pruning and quantization shrink them, but they are typically studied in isolation. This work studies how these methods behave and how they interact when applied to Multi-Layer Perceptrons (MLPs) on two MedMNIST datasets, PneumoniaMNIST and BloodMNIST. I first establish classical and MLP baselines, then apply pruning, quantization, coreset selection, and PCA individually, and finally study three interactions: the order of pruning and quantization, the effect of batch size before and after compression, and combined compression. The MLP matches or beats the classical models, can be pruned to 90% sparsity and quantized to 3 bits with negligible loss, and PCA sharply reduces dimensionality, in one case improving accuracy. Order matters on the simpler task but not the harder one; the best batch size depends on the dataset, not compression; and naive combined compression collapses on the harder task. As an improvement, quantization-aware training (QAT) recovers up to 0.21 accuracy in the failure cases and reduces variance, showing that QAT is necessary, not optional, for stable combined compression.

1 Introduction

Neural networks are often larger than they need to be which makes them slow to run and heavy to store. Compression methods like pruning (removing weights) and quantization (storing weights with fewer bits) shrink them but most work studies these methods one at a time. The goal of this project is to study how these methods behave and how they interact when applied to MLPs. I implement each compression method on its own, compare classical machine learning against the MLP, and then study the interactions. The research questions driving the project are: (1) does the order of pruning and quantization matter; (2) does the best batch size shift once a model is compressed; and (3) does combining methods beat using any one alone?

2 Related Work

Pruning has a long history beginning with second-order methods such as Optimal Brain Damage [1], which removes weights using curvature information. Modern magnitude pruning was popularised by Han et al. [2], who showed that removing low-magnitude weights and retraining preserves accuracy. The Deep Compression pipeline [3] combined pruning with quantization and Huffman coding to achieve large size reductions. The Lottery Ticket Hypothesis [4] showed that sparse trainable subnetworks exist inside dense ones. For batch size, Keskar et al. [5] showed that large-batch training tends to converge to sharp minima that generalise worse, and Smith et al. [6] showed batch size and learning rate are partly interchangeable. The datasets used here come from the MedMNIST benchmark [7], which provides standardised lightweight medical image classification tasks. Most of these works study a single compression axis; this project instead examines how several interact.

3 Methodology

Classical models: Three classical methods serve as comparison points: Logistic Regression, Random Forest, and an RBF-kernel SVM, each taking the flattened pixel vectors directly. These indicate whether a neural network is worth using.

MLP: The neural model is a 3-layer MLP (input $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ output) with ReLU activations and dropout, trained with Adam and cross-entropy loss. Validation accuracy is tracked each epoch, the best version is kept, and training stops early when it stops improving. This MLP is the model all compression methods are applied to.

Pruning: I use magnitude pruning, ranking all weights and setting the smallest fraction to zero, sweeping sparsity from 10% to 90%. Accuracy is measured immediately (raw damage) and after a short fine-tuning step that lets the surviving weights re-adjust and recover.

Quantization: I use simulated post-training quantization, where each weight is rounded to one of 2^b evenly-spaced levels between the smallest and largest weight, sweeping the bit-width b from 8 down to 2.

Coreset selection: The MLP is trained on random subsets of the data (25%, 50%, 100%) to test how much data is needed.

Dimensionality reduction: PCA is applied to the input before the MLP, sweeping the number of retained dimensions (16, 32, 64, 128).

Quantization-aware training (QAT): For the improvement, the model simulates quantization on every forward pass using a straight-through estimator, so it learns weights robust to low-precision rounding. QAT is applied to the pruned model.

4 Experiments

All experiments use a fixed seed (42) for reproducibility. The MLP uses Adam with weight decay 10^{-4}, batch size 128, dropout 0.5 (Pneumonia) or 0.3 (Blood) and learning rate 3×10^{-4} (Pneumonia) or 10^{-3} (Blood), with early stopping on validation accuracy. The MLP baseline and the QAT improvement are reported as mean $\pm$ standard deviation over 3 seeds (42, 43, 44); other sweeps are single-run. Pneumonia uses heavier regularisation because its test split is drawn from a different part of the source data than its training split, creating a natural accuracy ceiling. Pruning fine-tuning uses 10 epochs at learning rate 5×10^{-4}. Experiments were run in Google Colab. Evaluation metrics are test accuracy and macro-F1; macro-F1 is reported because PneumoniaMNIST is imbalanced. Datasets are summarised in Table 1.

Table 1: Dataset summary. Two MedMNIST datasets of differing modality and class count.

	PneumoniaMNIST	BloodMNIST
Type	Grayscale	RGB
Classes	2	8
MLP input size	784	2352
Train / Val / Test	4708 / 524 / 624	11959 / 1712 / 3421
Balance	Imbalanced	Mostly balanced

5 Results

5.1 Classical vs MLP baselines

On Pneumonia all models land around 85% and none clearly wins, due to the dataset's train/test source difference. On Blood the MLP wins, pulling ahead on cleaner, harder multi-class data.

Table 2: Classical methods vs MLP (test accuracy / macro-F1). MLP accuracy is mean $\pm$ std over 3 seeds. The MLP wins clearly on Blood.

Model	Pneu Acc	Pneu F1	Blood Acc	Blood F1
Logistic Regression	0.832	0.799	0.786	0.763
Random Forest	0.853	0.829	0.838	0.810
RBF SVM	0.857	0.832	0.853	0.832
MLP	**0.858 $\pm$ 0.002**	**0.833**	**0.867 $\pm$ 0.002**	**0.842**

5.2 Pruning

Blood drops sharply before retraining (0.478 at 90%) but recovers fully (0.874), showing the network is roughly 90% redundant. Pneumonia is robust throughout.

Table 3: Pruning: test accuracy before and after fine-tuning. Even at 90% sparsity, accuracy recovers to baseline or better.

Sparsity	Pneumonia		Blood	
	Before	After	Before	After
50%	0.849	0.867	0.865	0.873
70%	0.841	0.861	0.863	0.871
80%	0.832	0.870	0.779	0.870
90%	0.838	0.846	0.478	0.874

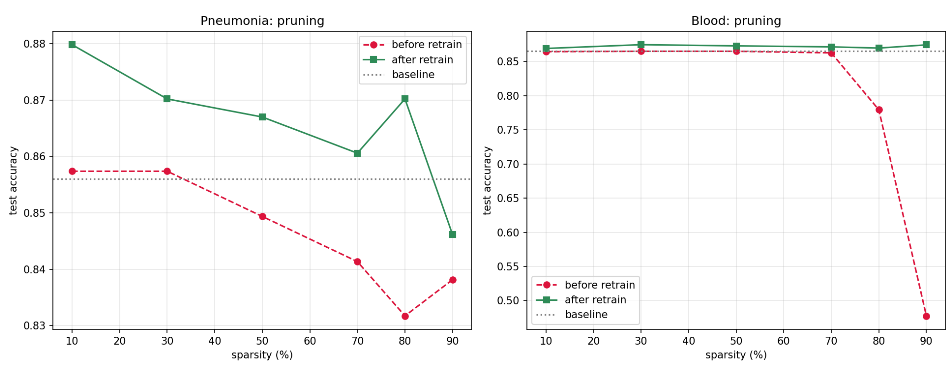

Figure 1: Pruning accuracy before (red) and after (green) fine-tuning, versus sparsity. The sharp drop at high sparsity is fully recovered after a short retraining step, showing the networks are highly redundant.

5.3 Quantization

Post-training quantization is lossless down to 3 bits on both datasets, but collapses at 2 bits on Blood (0.718), while Pneumonia degrades only slightly. The harder multi-class task has less margin to absorb the coarse 2-bit rounding.

Table 4: Quantization: test accuracy vs bit-width. Lossless to 3 bits, breaks at 2.

Bits	Pneumonia	Blood
32 (full)	0.856	0.865
8	0.856	0.865
4	0.859	0.862
3	0.861	0.865
2	0.848	0.718

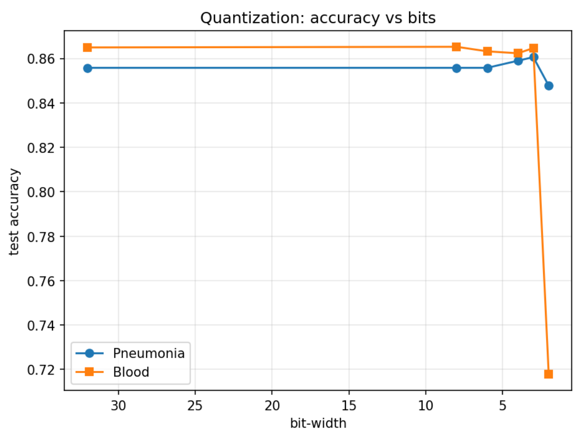

Figure 2: Quantization: test accuracy vs bit-width. Accuracy is flat down to 3 bits, then Blood drops sharply at 2 bits while Pneumonia holds.

5.4 Coreset and PCA

Coreset shows 50% of the data reaches most of the accuracy (0.803 at 25%, 0.842 at 50%, 0.860 at 100%). PCA reduces 784 inputs to 128 dimensions while improving Pneumonia accuracy to 0.883, because PCA removes noise; Blood needs more dimensions (0.852 at 128) since RGB images carry more information.

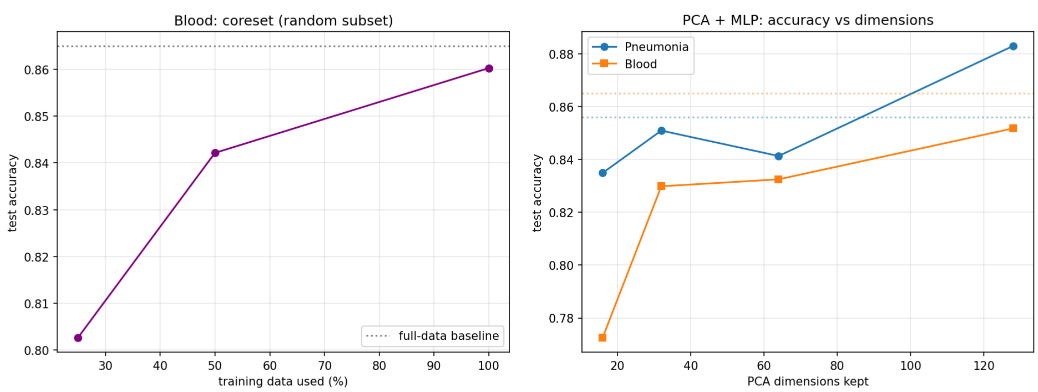

Figure 3: Left: coreset, accuracy vs fraction of training data (Blood); 50% of the data already reaches most of the accuracy. Right: PCA, accuracy vs retained dimensions; Pneumonia improves at 128 dimensions as PCA removes noise.

5.5 Interaction 1: Order of pruning and quantization

Using a protocol that fine-tunes after every step, I compare prune-then-quantize (P→Q) against quantize-then-prune (Q→P) at matched compression. On Pneumonia, P→Q is consistently better, by up to +0.19 at aggressive settings. On Blood, at sensible compression levels (4–6 bit, $\leq$70% sparsity) the two orders are equivalent (within $\pm$0.01), and at very aggressive settings the combined compression becomes unstable regardless of order. Order therefore matters more on the simpler task; the harder multi-class task has less spare capacity and is dominated by compression severity rather than order.

Table 5: Order interaction (test accuracy). Prune-then-quantize is consistently better on Pneumonia; neutral on Blood at sensible levels.

Setting	P→Q	Q→P	Diff
Pneu 90%/3-bit	0.853	0.796	+0.057
Pneu 70%/2-bit	0.851	0.665	+0.186
Blood 70%/4-bit	0.866	0.871	-0.005
Blood 50%/6-bit	0.870	0.873	-0.004

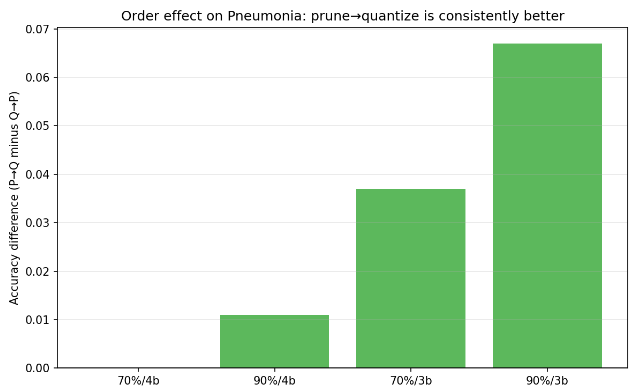

Figure 4: Order effect on Pneumonia: prune-then-quantize minus quantize-then-prune. The advantage grows as compression becomes more aggressive.

5.6 Interaction 2: Batch size and compression

The best batch size does not shift after compression: the optimum for the full model is also the optimum for the compressed model. However it is dataset-dependent, Pneumonia prefers a large batch (256), Blood a small one (64). This matches Keskar et al. [5]: the harder task is more sensitive to large batches. The compressed model also slightly outperforms the uncompressed one at every batch size, consistent with compression acting as a regulariser.

Table 6: Best batch size: uncompressed vs compressed MLP (test accuracy).

	Pneumonia		Blood	
Batch	Uncomp.	Comp.	Uncomp.	Comp.
32	0.856	0.870	0.859	0.866
64	0.859	0.864	**0.866**	**0.873**
128	0.859	0.870	0.863	0.866
256	**0.873**	**0.875**	0.862	0.867

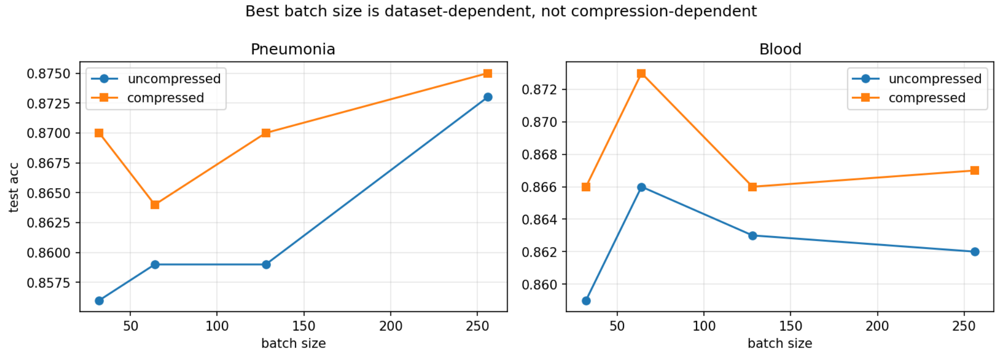

Figure 5: Best batch size, uncompressed vs compressed. The optimum is dataset-dependent (Pneumonia 256, Blood 64) and does not shift after compression; compressed slightly beats uncompressed everywhere.

5.7 Improvement: Quantization-Aware Training

Combining pruning with low-bit post-training quantization (PTQ) collapses on the harder task: Blood at 70% pruning with 2-bit falls to 0.602 with high variance. I replace PTQ with quantization-aware training (QAT), where the model simulates quantization during training, so it learns weights robust to low precision. Table 7 reports both over 3 seeds.

Table 7: Improvement: PTQ vs QAT under combined pruning and low-bit quantization (test accuracy, mean ± std over 3 seeds). QAT improves every configuration, with the largest gains where PTQ collapses, and lower variance.

Dataset	Prune	Bits	PTQ	QAT (ours)	Gain
Pneumonia	70%	3	0.847 ± 0.008	$\mathbf{0.867 \pm 0.007}$	+0.020
Pneumonia	90%	2	0.810 ± 0.026	$\mathbf{0.878 \pm 0.011}$	+0.068
Blood	70%	3	0.790 ± 0.020	$\mathbf{0.859 \pm 0.005}$	+0.069
Blood	90%	3	0.764 ± 0.008	$\mathbf{0.861 \pm 0.004}$	+0.097
Blood	70%	2	0.602 ± 0.083	$\mathbf{0.809 \pm 0.038}$	+0.207
Blood	90%	2	0.566 ± 0.029	$\mathbf{0.736 \pm 0.026}$	+0.170

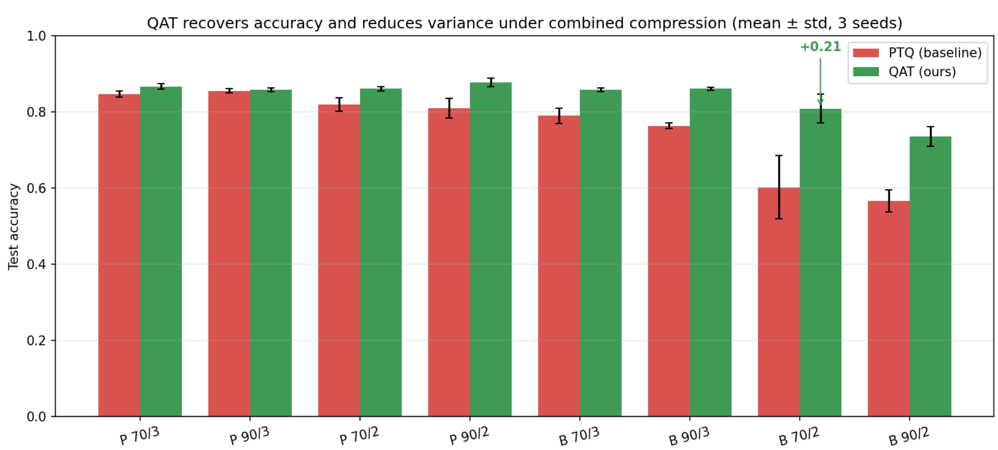

Figure 6: QAT (green) vs post-training quantization (red) under combined compression, mean ± std over 3 seeds. QAT improves every setting, recovers up to +0.21 where PTQ collapses, and has much smaller variance.

QAT improves every configuration. The largest gains occur where PTQ fails most: +0.207 on Blood at 70% pruning and 2-bit. QAT also reduces variance: PTQ's standard deviation reaches 0.083 in the failure cases, while QAT stays at 0.038 or lower. QAT is therefore both more accurate and more reproducible.

6 Discussion

Each compression method works well alone, but their interaction is more subtle. Three findings emerge. First, the order of pruning and quantization matters on the simpler binary task (prune-then-quantize is better, by up to +0.19 at aggressive settings) but not on the harder multi-class task at sensible compression; the order effect grows with compression severity. Second, the optimal batch size is a property of the dataset, not of compression, harder tasks prefer smaller batches, echoing prior large-batch findings. Third, naive combined compression breaks on the harder task, and quantization-aware training resolves this, recovering up to 0.21 accuracy and stabilising the result.

Limitations. The study uses MLPs, which discard spatial structure, so absolute accuracies sit below a CNN's; the focus is compression behaviour, not peak accuracy. Results span two datasets, and most compression sweeps are single-run, while the baseline and QAT improvement carry error bars over 3 seeds. Combined compression is unstable at extreme settings (2-bit), limiting the usable range.

Future work. Extending to more datasets and to CNNs would test whether the QAT-necessity finding generalises, and adding error bars across all sweeps would strengthen the statistical claims.

7 Conclusion

I studied how pruning, quantization, and batch size interact on MLPs across two MedMNIST datasets. Each compression method works well alone; their interaction is more nuanced. Order matters on simpler tasks, batch-size preference is dataset-driven, and naive combined compression fails on harder tasks. The main contribution is showing that quantization-aware training is necessary, not optional, for stable and accurate combined compression, recovering up to 0.21 accuracy and substantially reducing variance where standard post-training quantization collapses.

References

[1] LeCun, Y., Denker, J. S., & Solla, S. A. (1990). Optimal brain damage. In *Advances in Neural Information Processing Systems* (Vol. 2, pp. 598–605).

[2] Han, S., Pool, J., Tran, J., & Dally, W. (2015). Learning both weights and connections for efficient neural networks. In *Advances in Neural Information Processing Systems* (Vol. 28). `https://arxiv.org/abs/1506.02626`

[3] Han, S., Mao, H., & Dally, W. J. (2016). Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding. In *International Conference on Learning Representations (ICLR)*. `https://arxiv.org/abs/1510.00149`

[4] Frankle, J., & Carbin, M. (2019). The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations (ICLR)*. `https://arxiv.org/abs/1803.03635`

[5] Keskar, N. S., Mudigere, D., Nocedal, J., Smelyanskiy, M., & Tang, P. T. P. (2017). On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations (ICLR)*. `https://arxiv.org/abs/1609.04836`

[6] Smith, S. L., Kindermans, P. J., Ying, C., & Le, Q. V. (2018). Don't decay the learning rate, increase the batch size. In *International Conference on Learning Representations (ICLR)*. `https://arxiv.org/abs/1711.00489`

[7] Yang, J., Shi, R., Wei, D., Liu, Z., Zhao, L., Ke, B., Pfister, H., & Ni, B. (2023). MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1), 41. `https://www.nature.com/articles/s41597-022-01721-8`